Econ 131
Spring 2026

Midterm

March 2026

Student Name:

Student ID:

Exam Instructions

- **This exam has three questions. Each question is worth 50 points (150 points total). The total exam time is 75 minutes.**
- **Write answers using pens (we recommend black or blue ink), and write only in the space provided for each answer so your solutions scan properly.**
- **Do not write your solutions on pages that say "Do not write on this page"**. Answers written on these pages will not be graded.
- **Show your work**. Credit will only be awarded on the basis of what is written on the exam.
- **Affirm the academic honesty pledge below**. For those writing on a non-printed copy, please write "Academic Honesty Pledge as on exam" and sign your name. **If you do not affirm this pledge, your exam will be marked invalid.**

0. ACADEMIC HONESTY PLEDGE

I confirm that I have abided by all academic honesty rules for UC Berkeley and Economics 131. I confirm that I did not see this exam before my official exam start time. I confirm that I have not shared and will not share this exam with anyone else.

Signature: ______________________________

1. True/False/Uncertain Questions (50 points total)

You can start your answer "True", "False", "Uncertain", "Partially true", or "Partially false", etc., but your choice of the label does not matter much. What matters is your explanation.

(a) Redistribution is one reason for government intervention in the economy. *(10 points)*

(b) In a simple observational correlation, a large sample is enough to make the estimate causal. *(10 points)*

(c) If the GDP growth rate is higher than the interest rate on government debt, then the government can run a primary deficit while keeping debt/GDP stable. *(10 points)*

(d) If taxes and transfers have absolutely no behavioral effects, a utilitarian government sets taxes and transfers so that everyone consumes the same amount. *(10 points)*

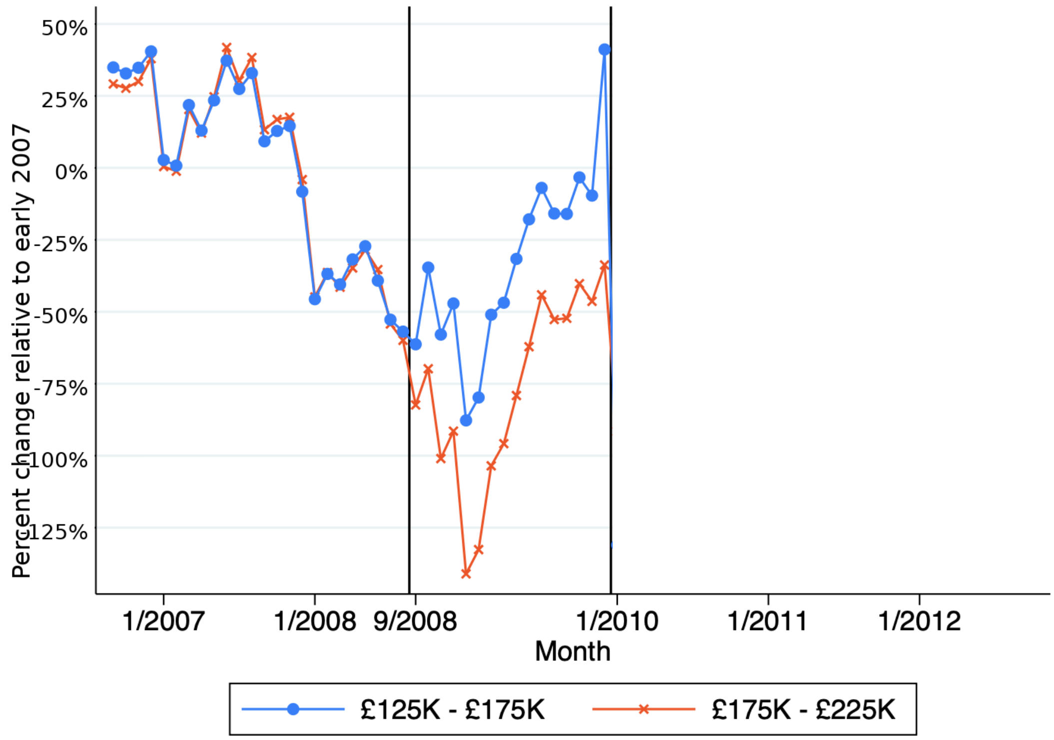
(e) An Earned Income Tax Credit is desirable when the intensive margin response is larger than the extensive margin response. *(10 points)*

2. Empirical Application: Best and Kleven (Housing Market Responses to Transaction Taxes) (50 points total)

In the the United Kingdom, the government taxes home sales. In resonse to the Great Recession, the UK government in Setptember 2008 eliminated the tax on homes in the sale price range between £125,000 and £175,000. This temporary tax break lasted for 16 months, from September 2008 through December 2009. The tax rate on properties in the sale price range between £175,000 and £225,000 remained unchanged.

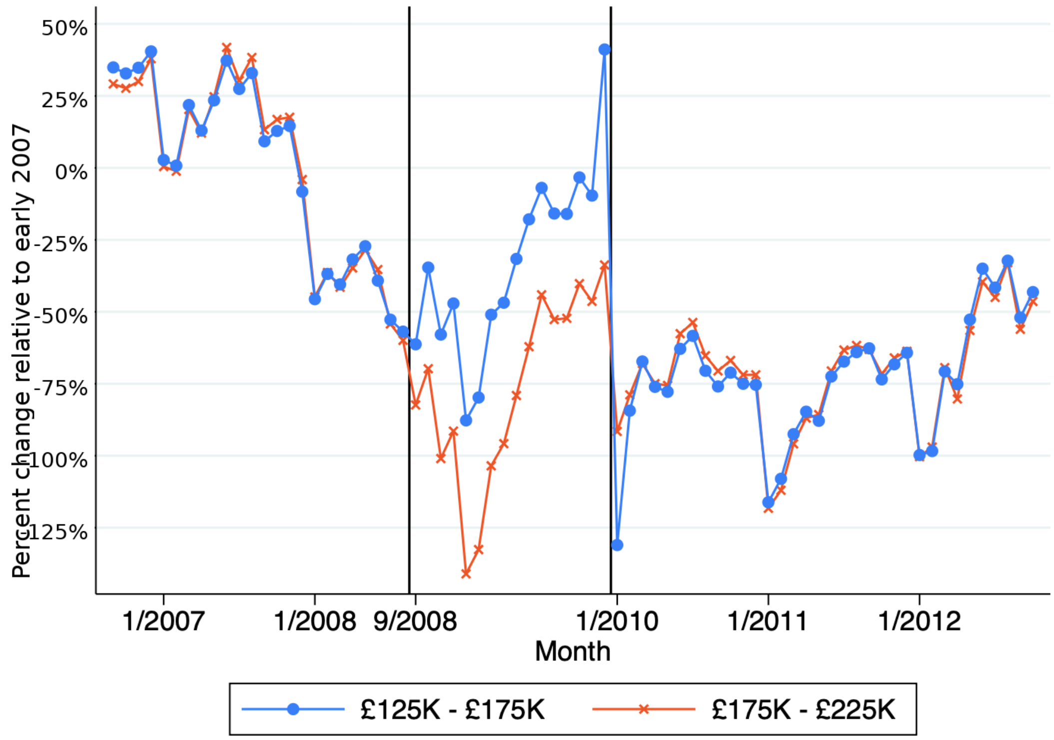
For all figures in this question, the y-axis ranges from -125% ***to*** 50%***.***

(a) Suppose you find that in the 16-month period, more homes were sold in the £125,000-£175,000 price range than in the £175,000-£225,000 price range. Is that good evidence that the tax break caused an increase in home sales in the £125,000-£175,000 price range, why or why not? *(12.5 points)*



(b) Suppose you draw the figure above. The vertical black line indicates September 2008. The circles show the number of home sales in the £125,000-£175,000 price range, expressed as the percent change from early 2007. The stars show the number of home sales in the £175,000-£225,000 price range, also expressed as the percent change from early 2007. Is this good evidence that the tax break caused an increase in home sales in the £125,000-£175,000 price range, why or why not? *(12.5 points)*

(c) Using at the graph above, approximately what do you think a difference-in-differences table would show is the difference-in-differences estimate of the causal effect of the tax break on home sales in the £125,000-£175,000 price range? (You may use the following numbers. The graph's pre-treatment mean was approximately 0% for both the £125,000-£175,000 group and for the £175,000-£225,000 group. The graph's post-treatment mean was approximately -25% for the £125,000-£175,000 group and approximately -75% for the £175,000-£225,000 group.) *(12.5 points)*



(d) Now suppose you extend the figure to include data points after December 2009, as in this second figure. Does this new figure provide additional comforting evidence that the tax break caused an increase in home sales in the £125,000-£175,000 price range, why or why not? *(12.5 points)*

3. Incidence of Commodity Taxation (50 points total)

Consider the following stylized market for televisions in Berkeley. Let market demand and supply be summarized as

$$Q^D = 912 - \frac{P}{5}, \qquad Q^S = \frac{P}{10},$$

where P is the dollar price per TV and Q is measured in thousands of TVs.

(a) Solve for the competitive equilibrium before any tax. Report equilibrium price and quantity. *(10 points)*

(b) Now impose a per-unit tax of $t = \$60$ on buyers. Solve for the post-tax equilibrium and report the consumer price, producer price, and quantity. *(10 points)*

(c) Compute the deadweight loss from the tax and illustrate it on a supply-demand graph. *(10 points)*

(d) Compute the tax incidence on consumers and producers, and explain intuitively which side bears more of the burden and why. *(10 points)*

(e) Suppose consumers are fully inattentive to the tax and keep using the original demand equation as if no tax exists. What does this imply for deadweight loss and incidence? You do not need to do any math; full credit will be given for clear intuition. *(10 points)*

Reminder: please sign the Academic Honesty Pledge on the first page!